

U.S. Fish & Wildlife Service

Lower Great Lakes Fish and Wildlife Conservation Office, Aquatic Invasive Species Program

Early Detection Monitoring Implementation Plan, Lake Erie and Lake Ontario 2025

Program Overview

Aquatic invasive species pose a serious threat to the Great Lakes. That threat has prompted natural resource agencies to examine pathways of introduction and risks posed by priority species. Prevention, early detection, and rapid response provide the most effective and potentially successful means to minimize harm to the environment and costs associated with stopping the spread of nonnative species. The U.S. Fish and Wildlife Service (USFWS) developed the *Strategic Framework for the Early Detection of Non-native Fishes and Select Benthic Macroinvertebrates in the Great Lakes* identifying early detection as a priority action. The framework defines how the USFWS will carry out early detection efforts in high priority locations (HPL). HPLs are selected based on their increased likelihood for invasion. Sampling focuses on harbors, rivers, and embayments of the Great Lakes as determined by a risk-based prioritization framework for AIS in the Great Lakes. Results from the risk-based prioritization framework produce a ranking for each taxa that represents invasion risk at a location and is used to select and prioritize survey locations (Table 1).

This report outlines implementation of the field elements of the Framework in Lake Erie and Lake Ontario during 2025. The proposed work will be carried out by the USFWS's Lower Great Lakes Fish and Wildlife Conservation Office (LGLFWCO).

Table 1: Priority levels of the top ranked locations across the lower Great Lakes.

Basin	Location	Ranking out of 386 Locations		Priority*
		Fish	Purpose	
Lake Erie	Buffalo, NY	3	Early Detection Monitoring	High
	Erie, PA	27		High
	North Tonawanda, NY	29		Moderate
	Upper Niagara River, NY/ON	54		Moderate
	Dunkirk, NY	76		Low
	Cattaraugus Creek, NY	94		Low
Lake Ontario	Rochester, NY	5		High
	Oswego, NY	6		High
	Sodus Bay, NY	57		Moderate
	Thousand Islands, NY/ON	71		Low
	Lower Niagara River, NY/ON	81		Low
	Oak Orchard Creek, NY	84		Low
	Massena, NY/ON	124		Low

* Priority designation was assigned based on rankings from the risk-based prioritization framework, supporting ongoing invasion fronts, and other AIS program objectives.

General Objectives

The objective of the framework is early detection; therefore, sampling strategies are designed to detect rare or low abundance species. Generally, sampling for rare species involves collecting the entire suite of species known to inhabit a location using a variety of gear types that sample multiple habitats and depth strata. The objective of this implementation plan is to execute a multi-gear sampling strategy that maximizes the potential for detecting a newly introduced fish species in a complex aquatic system while it is still rare and geographically restricted. This surveillance strategy uses both traditional (active and passive fisheries gear) and genetic (eDNA) methods to identify species presence across the various communities.

During the 2025 early detection monitoring field season, water bodies near the following locations will be surveyed (Figure 1): Buffalo, NY, Rochester, NY, Oswego, NY, Erie, PA, North Tonawanda, NY, Upper Niagara River, NY/ON, Sodus Bay, NY, Thousand Islands, NY/ON, Dunkirk, NY, Lower Niagara River, NY/ON, Oak Orchard Creek, NY, Cattaraugus Creek, NY, Massena, NY/ON, Braddock Bay, NY, Eighteen Mile Creek, NY, Salmon River Estuary, NY, Port Bay, NY, Little Sodus Bay, NY. The total proposed units of effort for juvenile/adult fish and metabarcoding sampling by each survey location can be found in Table 3. Gear will be used in accordance with the *Recommended Sampling Gear Types and Standard Operating Procedures for the Early Detection of Non-native Fishes and Select Benthic Macroinvertebrates in the Great Lakes* and will also follow safety and operation procedures implemented by the USFWS.

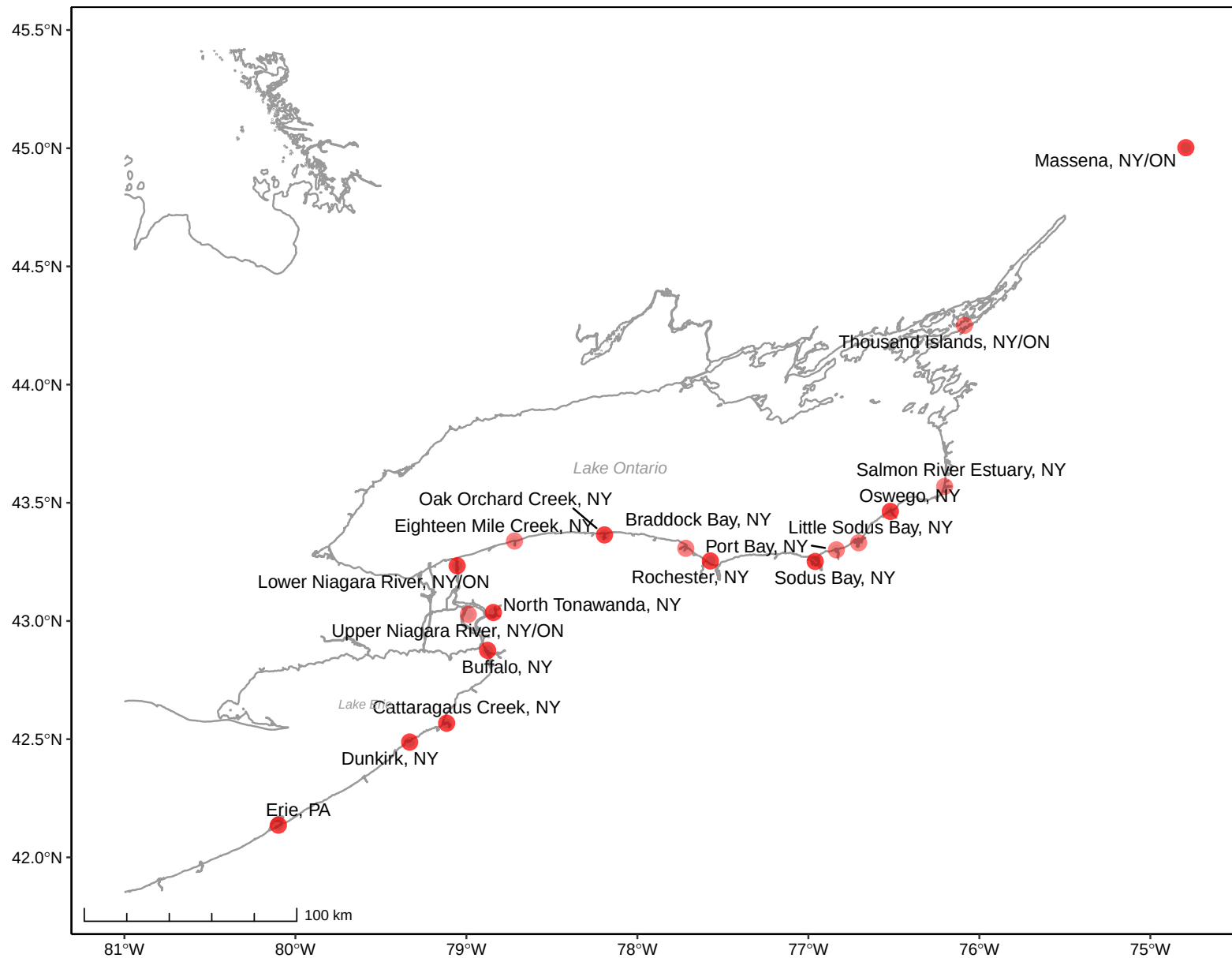


Figure 1: Locations for fish AIS surveys for the 2025 field season.

Traditional Surveillance Efforts

Starting in 2023, the USFWS Great Lakes Early Detection Monitoring program has undergone a review process that resulted in a standardized protocol for conducting broad-spectrum AIS surveillance at HPLs. This process has phases which focus on maintaining consistency in decision-making and gear allocation which will be analyzed to optimize adaptive sampling regimes for each location. Phase 1 describes a consistent 3-year sampling regime for new HPLs that have not previously been surveyed traditionally. None of the HPLs outlined in this plan fall into the phase 1 category, however, to build a baseline in which ongoing HPL surveillance can be analyzed under the protocol, a consistent 3-year sampling regime will be implemented from 2024-2026. During this 3-year sampling regime, a modified random site selection process that has 15% of sites in historically species rich habitats, as determined by historical data, will be implemented across all locations. Gear allocation for fish surveillance will be distributed roughly 80/20 between littoral and benthic habitats with gear allocation ratios being maintained during the 3-year sampling regime. Following the 3-year sampling regime, comparative analyses will be used to compare past sampling strategies. Metrics of success, which are still being developed through the review process, will also be applied to the 3-year sampling regime. Results from these analyses will be used to manipulate the site selection and gear allocation strategy to better optimize broad-spectrum AIS surveillance for each HPL.

Alongside the standard routine surveillance efforts of our early detection monitoring program, there are a few projects that will be undertaken that are worth mentioning:

1. Surveillance efforts along the St. Lawrence River near Massena, NY will be ongoing as Tench (*Tinca tinca*) continue to undergo a range expansion into the upper St. Lawrence River. This work is a product of a binational working group focused on tench detentions initiated through the Great Lakes Fish Commission. Partners include: New York State, St. Regis Mowhawk Tribe, DFO, McGill University, The River Institute, and Quebec Ministry of Natural Resources and Forests. This effort is targeted towards littoral habitats that have been suggested to be preferred by Tench and therefore will not be subscribed to the 3-year sampling regime. Massena, NY maintains a High priority status due to the range expansion of Tench and binational partner interest in monitoring that invasion front.
2. Although there will not be any dedicated crayfish or benthic invertebrate survey effort, further support of the Buffalo Park Schools Red Swamp Crayfish (*Procambarus clarkii*) infestation will result in a few days of effort around the perimeter of the known established population to determine any potential spread.

Genetic Surveillance Efforts

Environmental DNA (eDNA) is a molecular method of species detection that is used for AIS monitoring. The USFWS Great Lakes Early Detection Monitoring program has begun to incorporate multi-species eDNA (metabarcoding) monitoring into routine broad-spectrum surveillance efforts. As this new monitoring program gets implemented, objectives will include addressing research needs and beginning routine monitoring at HPLs. The LGLFWCO's Early Detection Monitoring eDNA objectives for 2025 are detailed below:

Early Detection Monitoring

Begin to implement eDNA metabarcoding as a surveillance method for novel AIS at lower priority HPLs where traditional surveillance is not currently occurring. This work will be supported through partnerships with the New York Department of Environmental Conservation (NYSDEC) and the USFWS LGLFWCO Grass Carp Strike Team.

St. Lawrence River Tench

Supporting Tench range expansion objectives in the St. Lawrence River using eDNA metabarcoding to detect Tench DNA above the final lock and dam upstream to Lake Ontario. This work will be supported through partnerships with Department of Fisheries and Oceans Canada (DFO) and the USFWS Ecological Services New York Field Office (NYFO).

The amount of eDNA sample effort for each of the projects listed above can be found in table 2. Metabarcoding effort associated with routine broad-spectrum surveillance at lower priority HPLs are also outlined in Table 3.

Table 2: Number of eDNA samples that will be allocated to each project.

Basin	Location	Purpose	Priority	Sample #*	Target Species
Lake Erie	Buffalo, NY	Early Detection Monitoring	High	20	
	Erie, PA		High	20	
	North Tonawanda, NY		Moderate	10	
	Upper Niagara River, NY/ON		Moderate	20	
	Dunkirk, NY		Low		
	Cattaraugus Creek, NY		Low		
	Buffalo, NY	Targeted	High	10	Grass Carp
	Erie, PA		Moderate		Grass Carp
	North Tonawanda, NY		High	10	Grass Carp
	Dunkirk, NY		Moderate		Grass Carp
	Cattaraugus Creek, NY		High		Grass Carp
Lake Ontario	Rochester, NY	Early Detection Monitoring	High	20	
	Oswego, NY		High	20	
	Sodus Bay, NY		Moderate		
	Thousand Islands, NY/ON		Low		
	Lower Niagara River, NY/ON		Low		
	Oak Orchard Creek, NY		Low		
	Massena, NY/ON		Low		
	Rochester, NY	Targeted	High	10	Grass Carp
	Braddock Bay, NY		Low		Grass Carp
	Oswego, NY		Moderate	10	Grass Carp
	Sodus Bay, NY		Moderate	10	Grass Carp
	Lower Niagara River, NY/ON		Moderate	10	Grass Carp
	Oak Orchard Creek, NY		High	10	Grass Carp
	Massena, NY/ON		High	110	Eurasian Tench
	Eighteen Mile Creek, NY		High	10	Grass Carp
	Salmon River Estuary, NY		High	10	Grass Carp
	Port Bay, NY		Low		Grass Carp
	Little Sodus Bay, NY		Low		Grass Carp
Total	-	-	-	310	-

Table 3: Amount of effort proposed for each high priority location for Juvenile/Adult fish, Metabarcoding, and Benthic Invertebrates.

Basin	Location	Purpose	Fish Surveys			
			Electrofish	Fyke Net	Gill Net	Metabarcoding
Lake Erie	Buffalo, NY	Early Detection Monitoring	20	10	10	20
	Erie, PA		20	10	10	20
	North Tonawanda, NY		10	10	10	10
	Upper Niagara River, NY/ON		20	20		20
	Dunkirk, NY					
	Cattaraugus Creek, NY					
	Buffalo, NY	Targeted				10
	Erie, PA					
	North Tonawanda, NY					10
	Dunkirk, NY					
	Cattaraugus Creek, NY					
Lake Ontario	Rochester, NY	Early Detection Monitoring	30	15	15	20
	Oswego, NY		20	5	15	20
	Sodus Bay, NY					
	Thousand Islands, NY/ON					
	Lower Niagara River, NY/ON					
	Oak Orchard Creek, NY					
	Massena, NY/ON					
	Rochester, NY	Targeted				10
	Braddock Bay, NY					
	Oswego, NY					10
	Sodus Bay, NY					10
	Lower Niagara River, NY/ON					10
	Oak Orchard Creek, NY					10
	Massena, NY/ON		60			110
	Eighteen Mile Creek, NY					10
	Salmon River Estuary, NY					10
	Port Bay, NY					
	Little Sodus Bay, NY					
Total	-	-	180	70	60	310